



Experimental and kinetic study on the laminar burning velocities of NH_3 mixing with CH_3OH and $\text{C}_2\text{H}_5\text{OH}$ in premixed flames

Zhihua Wang, Xinlu Han, Yong He*, Runfan Zhu, Yanqun Zhu, Zhijun Zhou, Kefa Cen

State Key Laboratory of Clean Energy Utilization, Zhejiang University, Zheda Road 38#, Zhejiang, Hangzhou 310027, P.R. China



ARTICLE INFO

Article history:

Received 28 November 2020

Revised 23 February 2021

Accepted 24 February 2021

Keywords:

Laminar burning velocity

Ammonia

Kinetic mechanism

Methanol

Ethanol

ABSTRACT

Ammonia (NH_3) is a promising carbon-free alternative fuel, and blending NH_3 with other more active fuels can enhance the low burning rate of pure NH_3 flames, which makes it more compatible with existing combustors. There are laminar burning velocity data of NH_3+H_2 , NH_3+CO /syngas, and NH_3+CH_4 flames in literature, however, blending NH_3 with the oxygenated alternative fuel alcohols have not yet been checked. In the present work, the laminar burning velocities of $\text{NH}_3+\text{CH}_3\text{OH}/\text{air}$ and $\text{NH}_3+\text{C}_2\text{H}_5\text{OH}/\text{air}$ flames were measured using the heat flux method at 1 atm with varied equivalence ratios and mixing ratios. Measurements were carried out at 298 K and elevated temperatures until 448 K unburnt temperatures, in order to check the data consistency via the temperature coefficient α in $\frac{S_L}{S_L^0} = (\frac{T_u}{T_u^0})^\alpha$. A new updated kinetic mechanism, based on our previous mechanism named CEU- NH_3 , was proposed and validated based on the present and published experimental data for laminar burning velocity, ignition delay times and species profiles. This mechanism contains 91 species and 444 reactions, which is small in size and thus advantageous to be used in the modeling of complex combustion fields. By numerical analyses using the CEU- NH_3 and the detailed Konnov mechanisms, the interactions between C- and N- containing species are found to be insignificant for the laminar burning velocities of the various NH_3 blending mixtures.

© 2021 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

1. Introduction

Recently, as a type of carbon-free fuel, ammonia (NH_3), has caught more and more attention by the combustion community. It can be produced from H_2 by the Haber Bosch process using renewable energy, e.g., solar and wind power, and has matured application in storage and transportation owing to the flourishing of the fertilizer industry. Though using NH_3 instead of the traditional hydrocarbon fossil fuels avoids the emission of greenhouse gas CO_2 , there are two main concerns regarding the industrial utilization of NH_3 as fuels, i.e., (1) the NO_x emissions, and (2) the low reactivity (relatively high resistance to ignition and low burning intensity). For the first issue, strategies like staged combustion have been proved successful in reaching low NO_x emissions in gas turbines [1,2]. As for the second issue, blending NH_3 with other fuels as additives can enhance the combustion properties, making the mixtures compatible with existing combustion equipment, e.g., hydrogen (H_2), CO, syngas, methane (CH_4), dimethyl ether (DME), and diesel, etc. [3–7].

Laminar burning velocity is one of the most important combustion characteristics in the fundamental research field. Up to now, there are NH_3+H_2 , NH_3+CO /syngas, and NH_3+CH_4 laminar burning velocity data available in the literature (e.g., [3,4,7]), however, the blending of NH_3 with more complex fuels are not well investigated. Among these fuels, the alcohols could be produced from bio-resources, thus having the nature of carbon-neutral advantages, and they have been investigated extensively in especially the blending with gasoline as oxygenated alternative fuels [8,9]. It is therefore of research interests to see what will happen if the carbon-free NH_3 is blended with the carbon-neutral alcohols, such as methanol (CH_3OH) or ethanol ($\text{C}_2\text{H}_5\text{OH}$), which will be helpful to develop a carbon-neutral energy system.

The simulation and detailed understanding of the combustion (or ignition) of the NH_3 blended mixtures rely on the kinetic mechanisms. Several mechanisms have been published targeting at NH_3 fueled systems. In chronological order, there are mechanisms from Mathieu and Petersen [10], Song et al. [11], Nakamura et al. [12], and Otomo et al. [13] for the NH_3+H_2 systems; a mechanism from our previous publication [4] for the NH_3+CO /syngas system; and mechanisms from Tian et al. [14], Okafor et al. [7], Shrestha et al. [15], and San Diego [16] for the NH_3+CH_4 (or higher-order hydrocarbon fuels) system, each of the latter can also work for

* Corresponding author.

E-mail address: heyong@zju.edu.cn (Y. He).

the former systems. From the NH_3+H_2 to NH_3+CH_4 system, the number of reactions and species increase rapidly due not only to the complexity of hydrocarbon reactions but also the interactions between C- and N-containing species, which makes the computational costs grow fast and difficult to be directly used for turbulent flame simulations. Besides the size of the mechanisms, their accuracy in predicting various fundamental combustion characteristics is also the main concern. For example, in the recent shock tube ignition delay time study of NH_3+CH_4 +air mixtures by Xiao et al. [17], the experimental results can be well predicted by the Tian et al. mechanism [14], which also reproduces the species profiles in low pressure burner stabilized NH_3+CH_4 flames [14]. However, when using the Tian et al. mechanism [14] to calculate the laminar burning velocities of NH_3+CO +air flames, large discrepancies are found comparing with the experimental data [3]. The mechanism in our previous publication [4] is small in size and has been proved to predict better the NH_3+H_2 , NH_3+CO , and NH_3 +syngas systems in general than the other mentioned mechanisms, thus to extend it carefully to be compatible with hydrocarbon oxidation has the potential to obtain a compressed and accurate kinetic mechanism, which is beneficial for optimization and improved design of NH_3 utilization systems.

Based on the background mentioned above, the aims of the present work are: (1) to measure the laminar burning velocities of NH_3 blended with different fuels including CH_3OH and $\text{C}_2\text{H}_5\text{OH}$ at varied mixing ratios and equivalence ratios at 1 atm using the heat flux method; (2) to develop and validate a more refined kinetic mechanism with relatively small size but accurately predict the present and literature published experimental data related with NH_3 combustion. It has been pointed out by Konnov and coworkers [18,19] that the measuring of laminar burning velocities at different unburnt temperatures can help identify the experimental data consistency, thus measurements will be made at 298 K, 348 K, 398 K, and 448 K unburnt temperatures; data of NH_3+H_2 and NH_3+CH_4 mixtures will also be measured to supplement the earlier 298 K laminar burning velocities published in [3].

2. Experimental details

In the present experiment, the laminar burning velocities S_L of $\text{NH}_3+\text{H}_2/\text{CH}_4/\text{CH}_3\text{OH}/\text{C}_2\text{H}_5\text{OH}$ +air flames were measured at atmospheric pressure using the heat flux method. The same method has been used and explained in our previous publications [3,4,20], thus there is only a brief introduction of the experimental setup. Flow rates of gas fuels and air (mixed by 21% O_2 and 79% N_2) were controlled by different mass flow controllers (MFC, Alicat). Since CH_3OH and $\text{C}_2\text{H}_5\text{OH}$ are in the liquid phase at normal condition, their flow rates are controlled by a liquid MFC, and a well-controlled evaporator mixer (CEM, Bronkhorst High-Tech) was used to vaporize and mix the liquid fuels with N_2 in the gas supply system. The arrangement of the gas supply system with CEM has been explained in detail in our previous publication about n-heptane and isooctane burning velocities [21], and a figure showing the setup can also be found in the present Supplementary material (see Fig. S1). The unburned mixture was then conducted to the heat flux burner, which measures the adiabatic laminar burning velocity S_L based on stretch-less, quasi-one-dimensional flat flames by changing the inlet velocity of the unburnt gas mixtures. The adiabatic state was checked by the temperature profiles measured by 8 thermocouples beneath the perforated burner plate of the heat flux burner. Uncertainty of the measured S_L mainly comes from the scatterings in MFCs and the temperature scatterings measured by thermocouples, as has been explained in our previous publications [3,4,20–22] as well as in literature studies [23,24]. For most of the measured S_L in the present study, the uncertainty evaluated is within ± 1 cm/s.

Table 1 shows the presently investigated experimental conditions. The Mixing ratio x_M refers to the mole fraction of fuel M in the total NH_3+M fuel mixture, where M could be H_2 , CH_4 , CH_3OH , and $\text{C}_2\text{H}_5\text{OH}$. The equivalence ratio ϕ is defined in such a way that the complete combustion products are N_2 , H_2O , and, if applicable CO_2 . For each fuel mixture, S_L data were measured at varied x_M , unburnt gas temperature T_u , and equivalence ratio ϕ . The mole fraction ratios among NH_3 , M, and air in the unburnt mixture is evaluated solely based on ϕ and x_M . Take $\text{M}=\text{CH}_3\text{OH}$ as an example, there is $x_{\text{CH}_3\text{OH}} \cdot \text{CH}_3\text{OH} + (1 - x_{\text{CH}_3\text{OH}}) \cdot \text{NH}_3 + 3.571 \cdot (1 + x_{\text{CH}_3\text{OH}})/\phi \cdot \text{air}$ in the mixture.

Data of 298 K NH_3+CH_4 and NH_3+H_2 have been measured and presented in 2019 [3], and they were repeated in the present study to see if there is data scattering when measuring at different period of time. It should be noted that at $T_u = 298$ K, experiments with large amount of CH_3OH or $\text{C}_2\text{H}_5\text{OH}$ can't reach the rich-side ϕ limit in the table because their saturated vapor pressures in this condition are not enough to keep the unburnt mixtures in the gas phase, which is required by the heat flux method. All the present S_L experimental data with uncertainties are tabulated in the Supplementary Material.

3. Kinetic modeling

3.1. Modeling details

Numerical simulations were carried out using ANSYS Chemkin 17.0 with the new proposed CEU-NH3 mechanism updated from our previous mechanism. The literature mechanisms from Shrestha [25], Glarborg [26], CRECK [27], and Konnov [28] are also used, which include all the fuel species used in the present experiments. The Shrestha mechanism is recently published has been validated using relevant conditions of ammonia combustion and ignition. Mechanisms from Glarborg CRECK, and Konnov are not aiming to model ammonia as a dominant reactant, however, they have detailed reaction schemes and it's difficult to find other mechanisms that include C2 alcohols and NH_3 at the same time, thus they were chosen here. The Konnov mechanism also includes the absorption coefficients of radiative species (NH_3 , NO , N_2O , CO , CO_2 , CH_4 , and H_2O), which were summarized from the work of Nakamura and Shindo [29]. The absorption coefficients can be used directly in Chemkin when the radiative heat losses need to be considered, which makes especially the S_L predictions more reasonable since the real experimental flames are subject to radiative heat losses [20,29]. Therefore, the absorption coefficients were manually added to other mechanisms used.

In the following, if the modeling data are to be compared with the experimental laminar burning velocity, the radiation heat losses were considered without further notification. Some data were calculated at adiabatic conditions for the analyses of adiabatic flame structures, and they will be noted as 'adiabatic' datasets. All the S_L simulations used the 'Flame_speed' module, and have taken into account the Soret and multi-component transportation effects. In the Supplementary Material, there will be comparisons between literature shock tube experimental data and modeling results, which were calculated using the 'Close homogenous' module, and the ignition delay times were derived from the criteria in the corresponding literature experiments.

3.2. CEU-NH3 mechanism

The present kinetic mechanism, named 'CEU-NH3' in the following, was updated from the NH_3 +syngas reaction scheme from our previous publication [4]. In [4], the NH_3 reactions have complied with the ELTE syngas mechanism [30]; the obtained reaction

Table 1
Experimental conditions investigated in the present study.

Fuel mixture	Mixing ratio x_M	Equivalence ratio ϕ	Unburnt Temperature T_u (K)	Notes
NH ₃ +H ₂	0.15, 0.25, 0.35, 0.4	0.8–1.4	298	Repeat [3]
			348, 398, 448	New data
NH ₃ +CH ₄	0.2, 0.4, 0.6, 0.8	0.7–1.5	298	Repeat [3]
			348, 398, 448	New data
NH ₃ +CH ₃ OH	0–1 0.2, 0.4, 0.6, 0.8, 1	1 0.7–1.8 ^a	298	New data
			298	
			348, 398, 448	
NH ₃ +C ₂ H ₅ OH	0–1 0.2, 0.4, 0.6, 0.8, 1	1 0.7–1.6 ^b	298	New data
			298	
			348, 398, 448	

^a Experiments containing a large amount of CH₃OH at 298 K cannot reach $\phi = 1.8$ because of the limit in the saturated vapor pressure of CH₃OH.

^b Similar to the CH₃OH conditions.

Table 2

Updated and added reactions in the present mechanism. Units are s, mol, cm, cal in $k = AT^n \exp(-\frac{E_a}{RT})$.

No.	Reaction	A	b	E _a	Notes	Ref.
1	N+NO=N ₂ +O	1.53E13	−0.0027	−185.41	Updated to improve the thermal NO performance in CH ₄ flames	[28]
2	T-CH ₂ +H=CH+H ₂	3.12E13	0	−1341	Updated to improve the CH and prompt NO performance in CH ₄ flames	[33]
3	CH+CO ₂ =HCO+CO	0.03	4.44	−3537	Updated with theoretical pressure-dependent rate constants to improve the S _t prediction of CH ₄ , CH ₃ OH, and C ₂ H ₅ OH flames	[34]
4	CH ₃ +OH=CH ₂ OH+H	1.62E10 PLOG/0.01 1.621E+10 0.965 3214.0/ PLOG/0.10 1.807E+10 0.950 3247.0/ PLOG/1.00 4.686E+10 0.833 3566.0/ PLOG/10.0 1.525E+13 0.134 5641.0/ PLOG/100 3.590E+14 −0.186 8601.0/ 7.4E10	0.965	3210		[38]
5	CH ₂ OH(+M)=CH ₂ O+H(+M)	LOW/3.5E21 −1.99 23,983 / TROE/ 0.844 900 1 3315 / HE/0.67/ AR/0.85/ O2/1/ H2/2/ CO/1.5/ CO2/2/ H2O/6/ CH4/2/ CH2O/2.5/ C2H6/3/ CH3OH/3/	0.811	39,559		[39]
6	CH ₃ CH ₂ O(+M)=CH ₃ +CH ₂ O(+M)	6.3E10 LOW/4.7E25 0.930 16,532/ TROE/0.426 0.3 2278 100,000/	0.93	17,098		[40]

scheme is small in size and predicts well the laminar burning velocity, ignition delay time, and the NO_x emission characteristics in the NH₃, NH₃+H₂, and NH₃+syngas systems. It should be noted that here 6 species (HONO, HNOH, HNO₂, HONO₂, NO₃, HON) and the corresponding 43 reactions could be deleted if N₂O is not the main component of reactants [4]. To make the CEU-NH₃ mechanism capable of predicting NH₃+hydrocarbon systems while still keeping relative small size, hydrocarbon reactions and interaction reactions between C- and N-containing species were selected respectively from the San Diego mechanism [16] and the recently updated Konnov mechanism [28].

The San Diego main mechanism can well predict the oxidation of hydrocarbon species until butane [31], where the numbers of species and reactions are kept to the minimum (57 species and 268 reactions in the 2016–12–14 released version) [16], thus its hydrocarbon oxidation part, i.e., reactions in the main mechanism excluding the syngas subset, were selected. There are NH₃ and NO_x sub-mechanisms available on the home page of San Diego, however, they are not considered in the present study because (1) the San Diego NH₃ subsets doesn't show better agreement with NH₃ experimental data as compared to the NH₃+syngas scheme from our previous publication [4]; and (2) the NO subset lacks the consideration about NCN reactions initiating the prompt NO formations due to the early time it was developed (2004).

The C–N interaction reactions were selected from the Konnov mechanism [28,28] was motivated by the findings of Brackmann et al. [32] that none of the literature mechanisms can well predict the measured NO mole fractions in atmospheric CH₄+air flames. In [28], reaction rates of the most important thermal NO reaction NO+N = N₂+O were derived from the experimental NO data with different oxygen ratio in air, and simulations about the CH, HCN, and NCN concentration in flames were carried out to make a comparison with literature experimental data and improve the predictions of prompt NO. The C–N interaction reactions in [28] were reduced to a subset with 12 species and 52 reactions before being added to the CEU-NH₃.

The thermodynamic and transport data were collected according to the selected species in each mechanism. The absorption coefficients of each radiative species were collected from the Konnov mechanism [28].

The above procedure is a pure compilation of hydrocarbon reactions from the San Diego main mechanism [16], the reduced C–N reactions from the Konnov mechanism [28], and the NH₃+syngas reaction scheme [4]. This procedure guarantees that the CEU-NH₃ mechanism is small in size (91 species and 444 reactions) and predicts well the systems without hydrocarbon fuels since the NH₃-syngas system validated by [4] is kept, however, it's not necessarily giving satisfactory predictions on the hydrocarbon fuels as well as

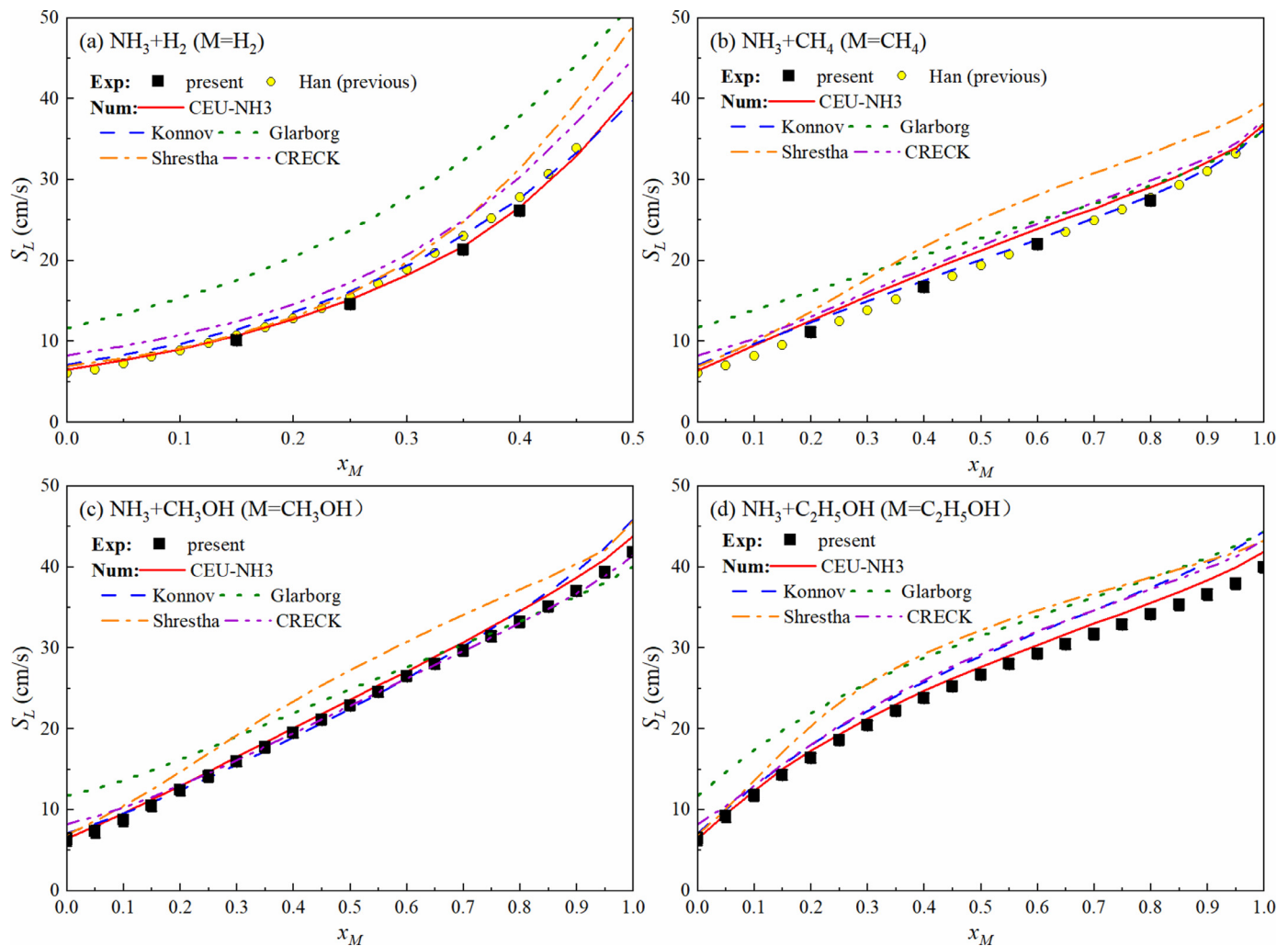


Fig 1. Laminar burning velocity of stoichiometric (a) NH_3+H_2 +air, (b) NH_3+CH_4 +air, (c) $\text{NH}_3+\text{CH}_3\text{OH}$ +air, and (d) $\text{NH}_3+\text{C}_2\text{H}_5\text{OH}$ +air flames, at 298 K and 1 atm.

the interactions between hydrocarbon and NH_3 . To make CEU-NH3 capable of describing these, S_L data of $\text{CH}_3\text{OH}/\text{C}_2\text{H}_5\text{OH}$ +air flames in present experiment and the NO formation and S_L of CH_4 +air flames in [28] were chosen as targets. By sensitivity analyses, 6 reactions were updated to meet the targets by using more recent rate constants, which are summarized in Table 2. Rate constants of the updated R1 were taken from [28], which help to reproduce better the NO mole fractions in the lean side of CH_4 +air and CH_3OH +air flames. The modifications of R1 have a negligible influence on the data validated in the NH_3 +syngas reaction scheme [4].



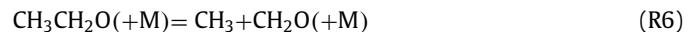
Rate constants of the updated R2 and R3 come respectively from Devriendt et al. [33] and Mehlmann [34], which improves the prompt NO performance through modifying the CH concentration in flames.



Though other high-sensitivity CH reactions have more recent reaction rates, they are not to be modified since the San Diego main mechanism is a reduced mechanism that doesn't include all possible reaction channels. For example, $\text{CH} + \text{O}_2 \rightarrow \text{product}$ reaction is estimated in literature [35,36] and other detailed mechanisms [26,37] with a few main product channels, while the San Diego

main mechanism considered only $\text{CH} + \text{O}_2 = \text{HCO} + \text{O}$, thus it's difficult and unreasonable to modify the only selected channel based on sources proving other main channels.

Rate constants of the updated R4–R6 are respectively from the theoretical computations by [38,39], and [40]. The updating of R4–R6 is not to compensate for the effect of substituting the syngas chemistry in the San Diego mechanism by the ELTE (in fact the substitution leads to less than 5% difference in, e.g., CH_4 laminar burning velocity), rather, these reactions make the predictions agree better with the targeted laminar burning velocity.



The comparison between the CEU-NH3 mechanism simulation results and various experimental data can be found in the Supplementary Material (Figs. S2–S6). These experimental data are: (1) laminar burning velocities of CH_4 , CH_3OH , $\text{C}_2\text{H}_5\text{OH}$ + air flames from [28,41–49] and present experiment; (2) NO mole fractions in $\text{CH}_4+\text{N}_2+\text{O}_2$ and CH_3OH +air flames from [28,32]; (3) ignition delay times of CH_4 , CH_3OH , and $\text{C}_2\text{H}_5\text{OH}$ mixtures from [50,51]; (4) species profile and ignition delay time data of the CH_4+NH_3 +air mixtures from [14,17]. Comparisons for the $\text{NH}_3+\text{CH}_3\text{OH}/\text{C}_2\text{H}_5\text{OH}$

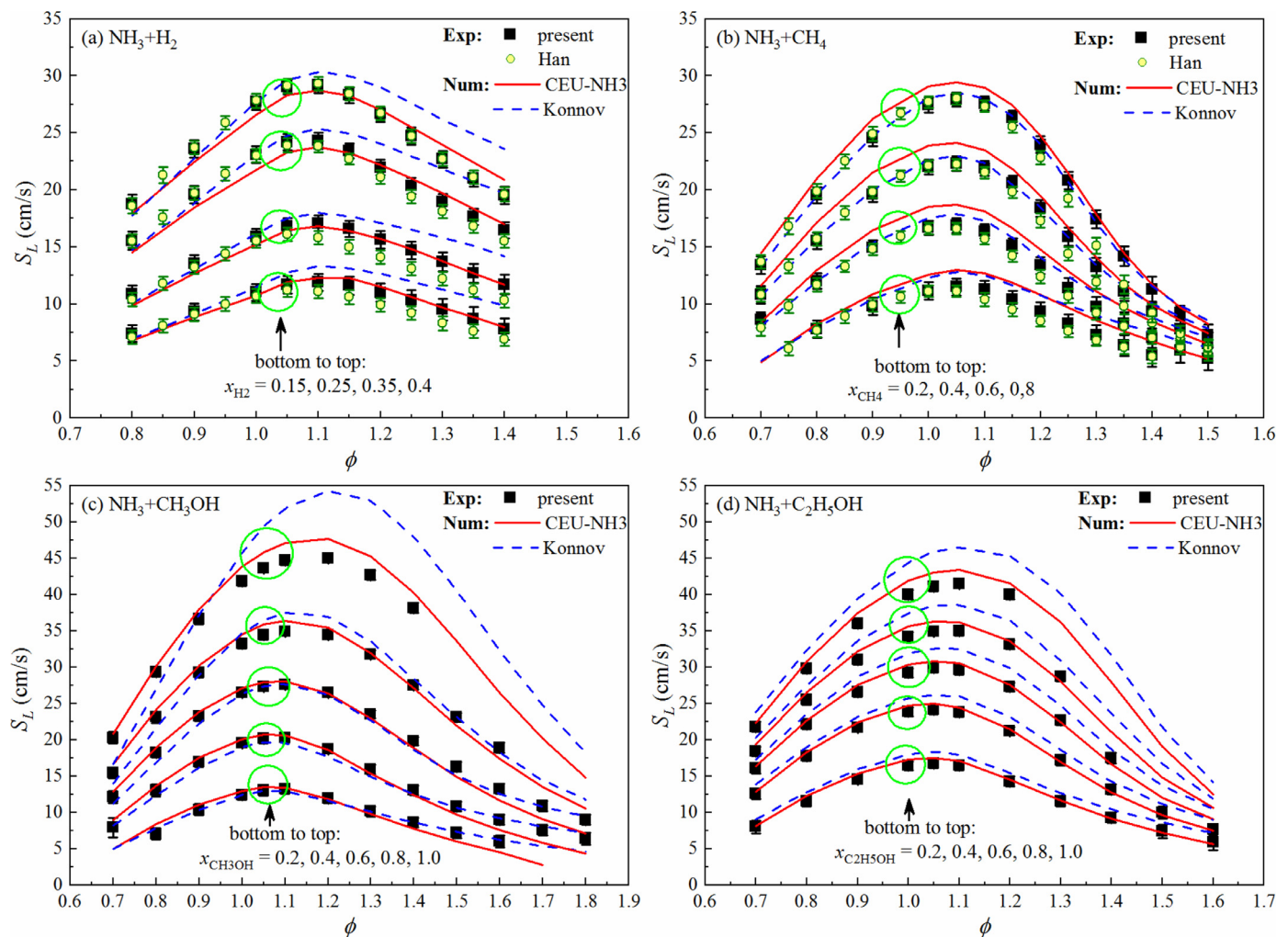


Fig 2. Laminar burning velocity of (a) $\text{NH}_3 + \text{H}_2 + \text{air}$, (b) $\text{NH}_3 + \text{CH}_4 + \text{air}$, (c) $\text{NH}_3 + \text{CH}_3\text{OH} + \text{air}$, and (d) $\text{NH}_3 + \text{C}_2\text{H}_5\text{OH} + \text{air}$ flames, at 298 K and 1 atm.

mixtures were not carried out due to the lack of literature experimental data. Overall, the CEU-NH3 can predict well for the tested systems. It should be noted that the CEU-NH3 doesn't include low-temperature reaction paths due to the limited size; also, the above validation shows no comparison with low-temperature conditions owing to the lack of experimental data. Therefore, the proposed CEU-NH3 mechanism is suggested to be used in the high-temperature conditions only.

4. Results and discussion

4.1. S_L at 298 K

Figure 1 shows the experimental and simulated laminar burning velocities of the stoichiometric $\text{NH}_3 + \text{H}_2/\text{CH}_4/\text{CH}_3\text{OH}/\text{C}_2\text{H}_5\text{OH} + \text{air}$ flames at 298 K and 1 atm, as a function of mixing ratio x_M . The $\text{NH}_3 + \text{H}_2$ and $\text{NH}_3 + \text{CH}_4$ data from the previous study by Han et al. [3] are included, which overlap with the present experimental results within the error bars for most cases investigated. This observation confirms the good repeatability of the experimental setup. In Fig. 1, the S_L increases with x_M , but the effect of adding different fuels are different, which will be further discussed using Fig. 3.

Figure 1 also shows the simulation results using 5 kinetic mechanisms. The Glarborg mechanism results are significantly larger than the experimental data especially at around the $x_M = 0$

conditions; the CRECK mechanism results show similar tendencies to the Glarborg, while the deviation from the experimental data is much smaller. The Shrestha mechanism well predicts the pure NH_3 S_L , however, it gives over-estimations at around $x_M = 0.5$, regardless of fuels. These performances indicates that these three mechanisms could be further improved regarding the present experimental conditions. From Fig. 1, the CEU-NH3 and Konnov mechanisms give the overall best predictions, and thus they will also be used in the following discussions.

Figure 2 shows the present experimental results of $\text{NH}_3 + \text{H}_2/\text{CH}_4/\text{CH}_3\text{OH}/\text{C}_2\text{H}_5\text{OH} + \text{air}$ laminar burning velocities at 298 K and 1 atm, as functions of equivalence ratio ϕ and mixing ratio x_M . In Fig. 2, and the equivalence ratio of maximum S_L is always found around $\phi = 1.05 - 1.1$. For the simulation results using the CEU-NH3 and the Konnov mechanisms, both mechanisms can reproduce the trends of the experimental data at varied x_M and ϕ , while none of them perfectly overlap with the experimental data. For $\text{NH}_3 + \text{H}_2$ flames, the CEU-NH3 is slightly lower than the experimental results at the fuel-lean side, while the deviations predicted by the Konnov mechanism grow larger at increasing ϕ . The CEU-NH3 mechanism results are higher by 1–2 cm/s for stoichiometric $\text{NH}_3 + \text{CH}_4$ flames, while this number decreases to be within 1 cm/s for the Konnov mechanism. As for the $\text{NH}_3 + \text{CH}_3\text{OH}$ and $\text{NH}_3 + \text{C}_2\text{H}_5\text{OH}$ flames, the Konnov mechanism simulation results are higher than the experimental data at the stoichiometric and fuel-rich side of flames, especially when

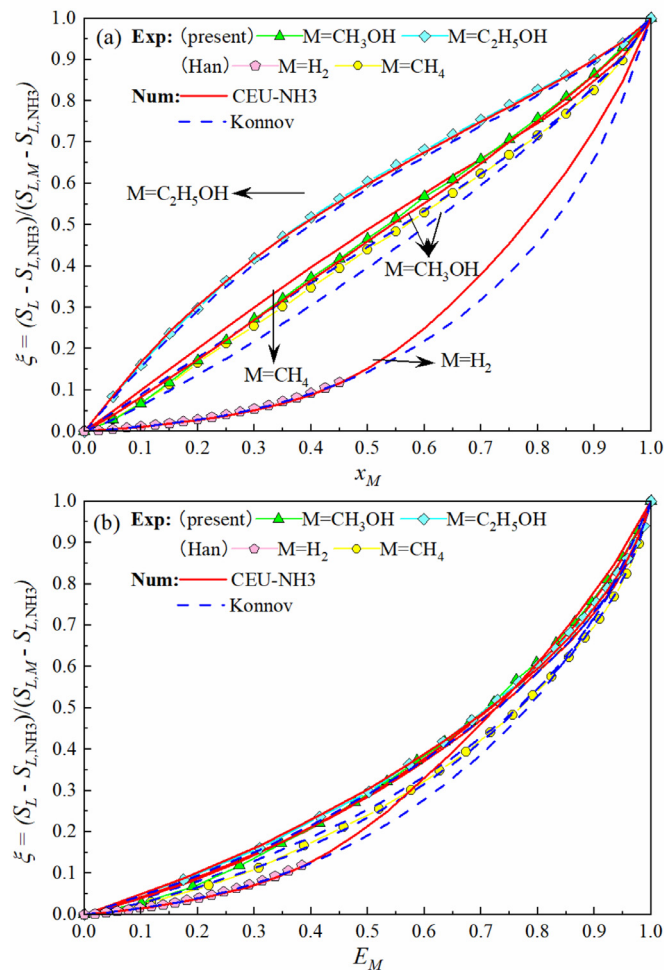


Fig. 3. Normalized laminar burning velocities at 298 K and 1 atm of stoichiometric $\text{NH}_3+\text{H}_2/\text{CH}_4/\text{CH}_3\text{OH}/\text{C}_2\text{H}_5\text{OH}+\text{air}$ flames as functions of (a) mole based fraction ratio x_M , and (b) energy fraction E_M .

the mixing ratio approaches the unity; however, better agreement with experimental data is achieved by the CEU-NH3 mechanism in these conditions.

Figure 3 show the same data as Fig. 1 in a slightly different way, in order to compare the effect of mixing different fuels to NH_3 flames. Normalizations have been made using the following parameter ξ ,

$$\xi = \frac{S_L - S_{L,\text{NH}_3}}{S_{L,M} - S_{L,\text{NH}_3}} \quad (1)$$

where S_{L,NH_3} and $S_{L,M}$ refer respectively to the laminar burning velocities at $x_M = 0$ and $x_M = 1$. By Eq. (1) there is always $\xi(x_M = 0) = 0$ and $\xi(x_M = 1) = 1$. The S_L of $x_{\text{H}_2} = 1$ flame is more than 200 cm/s, beyond the upper measurable limit of the heat flux method, thus the corresponding normalization of experimental results adopted the average of simulated S_{L,H_2} data. In Fig. 3, panels (a) and (b) are only different in the x-axis that the former uses the mole based mixing ratio x_M while the latter uses energy fraction E_M , which is defined as following,

$$E_M = \frac{x_M \cdot \text{LHV}_M}{x_{\text{NH}_3} \cdot \text{LHV}_{\text{NH}_3} + x_M \cdot \text{LHV}_M} \quad (2)$$

where LHV refers to the lower heating value. The energy fraction has been used by Okafor et al. [7] to investigate the laminar burning velocity of $\text{NH}_3+\text{CH}_4+\text{air}$ flames. From Fig. 3(a), the S_L data of mixtures with different M show different tendencies against

x_M : the NH_3+CH_4 and $\text{NH}_3+\text{CH}_3\text{OH}$ data almost overlap, showing a linear dependence on x_M ; while they are respectively higher and lower than the NH_3+H_2 and $\text{NH}_3+\text{C}_2\text{H}_5\text{OH}$ burning velocities at the same x_M . However, when converting x_M to E_M through Eq. (2), the four tested mixtures are found with same tendencies in Fig. 3(b), where the normalized burning velocities ξ almost overlap. The data tendency in Fig. 3(b) can be described by

$$\xi = E_M^\gamma \quad (3)$$

where the empirical γ parameter is around 2 fitted by the shown experimental and numerical datasets.

Based on these observations, it is clear that the S_L of the binary NH_3+M mixtures is mostly governed by the energy fractions of the mixing fuels regardless of what M is. Thus, it could be speculated that the interactions between NH_3 and M are not much different when M differs; and since NH_3+H_2 flames have no C–N interactions, this kind of interaction should not be important in NH_3+CH_4 , $\text{NH}_3+\text{CH}_3\text{OH}$, and $\text{NH}_3+\text{C}_2\text{H}_5\text{OH}$ flames, either. It should be noted that the above speculation comes from the analyses of the normalized S_L and the energy fraction, however, the same statement for the NH_3+CH_4 flames has been obtained in [3] from the reaction sensitivity analyses using the Okafor et al. [7] and the San Diego [16] mechanisms. Since the present work uses different mechanisms and extends the fuel types with CH_3OH and $\text{C}_2\text{H}_5\text{OH}$, reaction sensitivity analyses were also carried out.

Figure 4 shows the A-factor reaction sensitivities of laminar burning velocities shown in Fig. 3 at $x_M = 0.4$, calculated using the CEU-NH3 and Konnov mechanisms. In each panel, the bottom part shows the 10 most sensitive reactions among the entire mechanism, where none of the C–N interaction reactions exists. To find out how sensitive the C–N reactions could be, the 5 most sensitive C–N reactions were shown on the top of each panel with an asterisk. Although the chosen C–N reactions are not the same using the two mechanisms, the values of their sensitivities are negligible in both mechanisms shown in Fig. 4, proving true the proposed speculation in the last paragraph.

The above discussions confirm the ability of CEU-NH3 on predicting the $\text{NH}_3+\text{H}_2/\text{CH}_4/\text{CH}_3\text{OH}/\text{C}_2\text{H}_5\text{OH}+\text{air}$ flames, which is not affected by its compressed size as compared with the detailed Konnov mechanism. Therefore, in the following part, only the CEU-NH3 mechanism will be used to compare with the laminar burning velocities at elevated temperatures and derive the global reaction parameters.

4.2. S_L at elevated temperatures

Figure 5 shows the present experimental results of $\text{NH}_3+\text{H}_2/\text{CH}_4/\text{CH}_3\text{OH}/\text{C}_2\text{H}_5\text{OH}+\text{air}$ laminar burning velocities at 448 K and 1 atm, as functions of equivalence ratio ϕ and mixing ratio x_M . Plots of the experimental data at 348 K and 398 K with the simulation results using the CEU-NH3 mechanism can be found in Fig. S7 and Fig. S8 in the Supplementary material. Tendencies of the experimental data in Fig. 5 and Figs. S7–S8 are similar to that in Fig. 2. Simulation results using the CEU-NH3 mechanism are also shown in these plots, where there are discrepancies found in the experimental and simulation results for the NH_3+CH_4 flames and the $\text{NH}_3+\text{CH}_3\text{OH}$ flames at around stoichiometry, as compared with the discrepancies in Fig. 2.

Though the simulation results cannot perfectly overlap with the experimental data at elevated temperatures shown in Fig. 5, the discrepancies mentioned in the last paragraph indeed prove the data consistency in the present experiment, which can be analyzed in more detail by the temperature dependence coefficient α in $\frac{S_L}{S_L^0} = \left(\frac{T_L}{T_L^0}\right)^\alpha$ [52,53]. As suggested by Konnov and coworkers [18,19], α has much smaller reaction sensitivities than the S_L , making it nearly independent on the reaction mechanisms; thus the

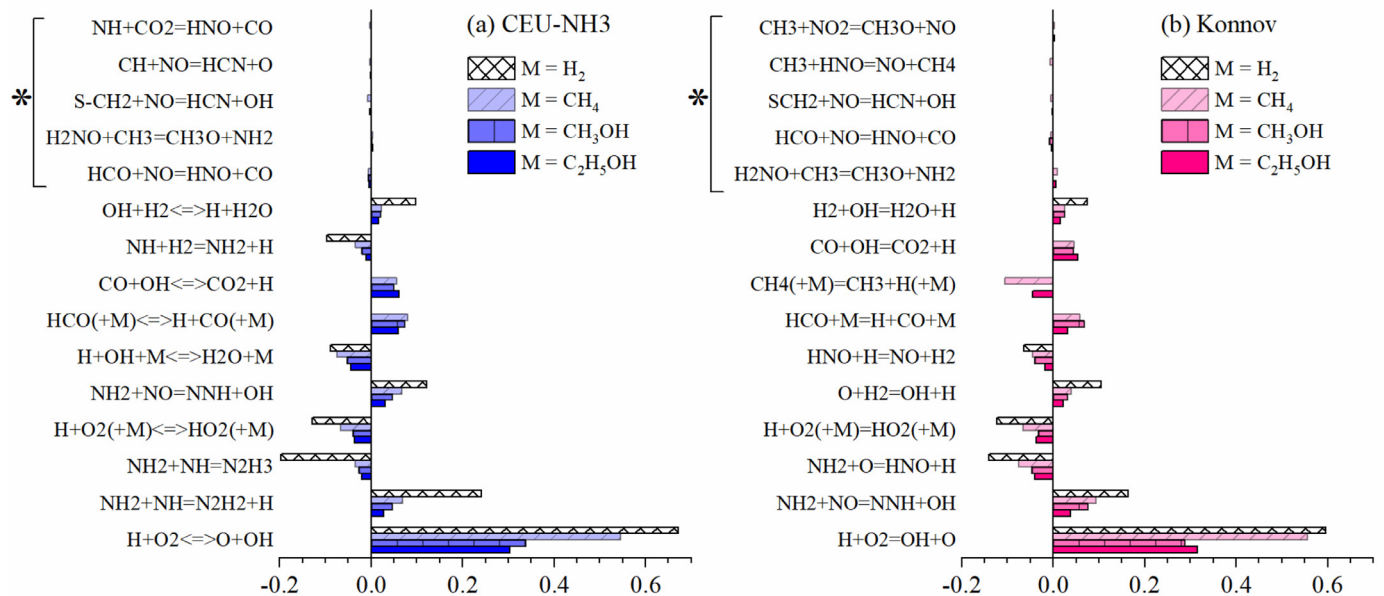


Fig. 4. A-factor reaction sensitivities of 298 K, 1 atm, and $x_M = 0.4$ stoichiometric $NH_3 + H_2/CH_4/CH_3OH/C_2H_5OH$ -air flames, using the (a) CEU-NH3 and (b) Konnov mechanisms.

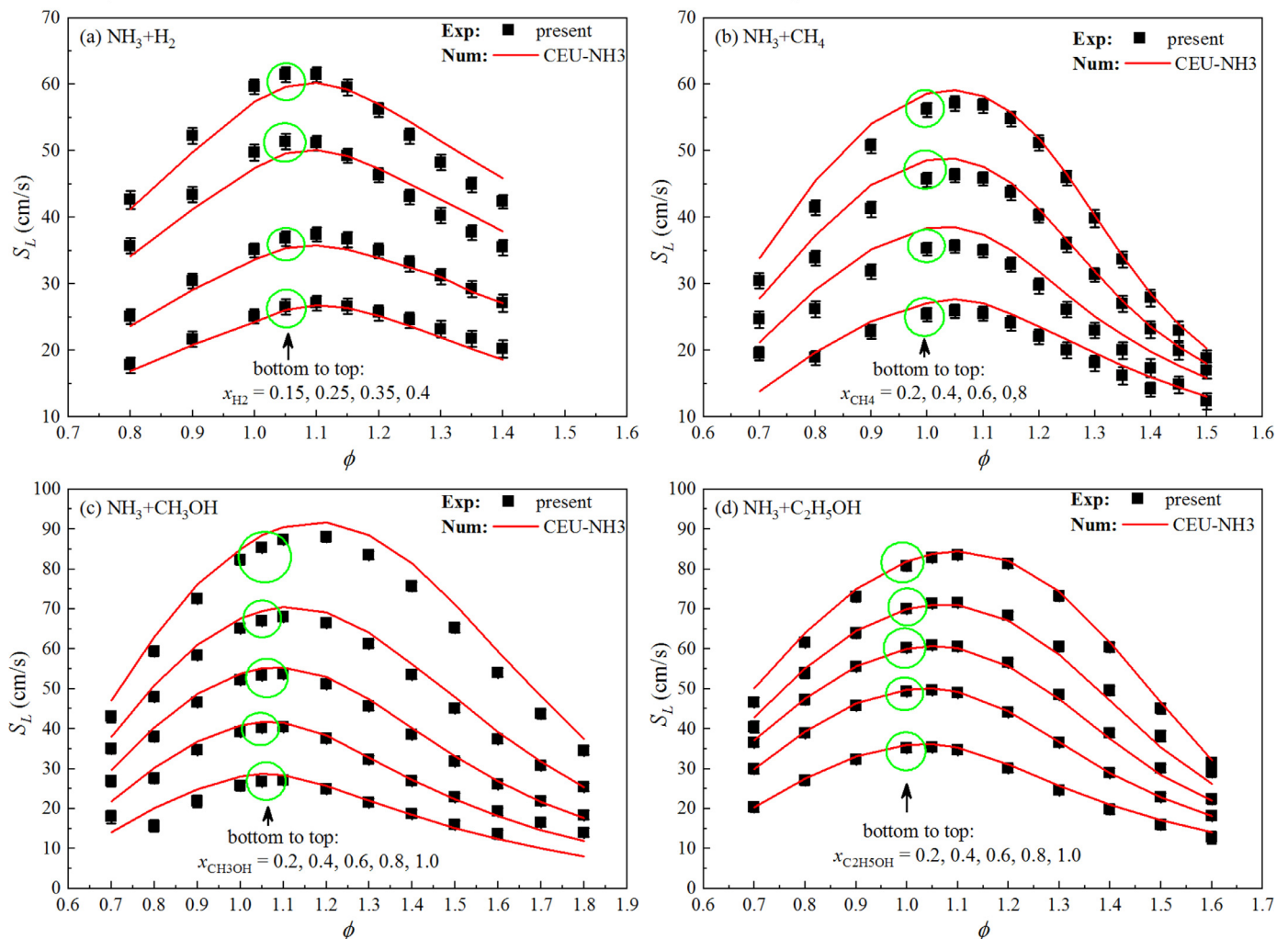


Fig. 5. Laminar burning velocity of (a) $NH_3 + H_2$ -air, (b) $NH_3 + CH_4$ -air, (c) $NH_3 + CH_3OH$ -air, and (d) $NH_3 + C_2H_5OH$ -air flames, at 448 K and 1 atm.

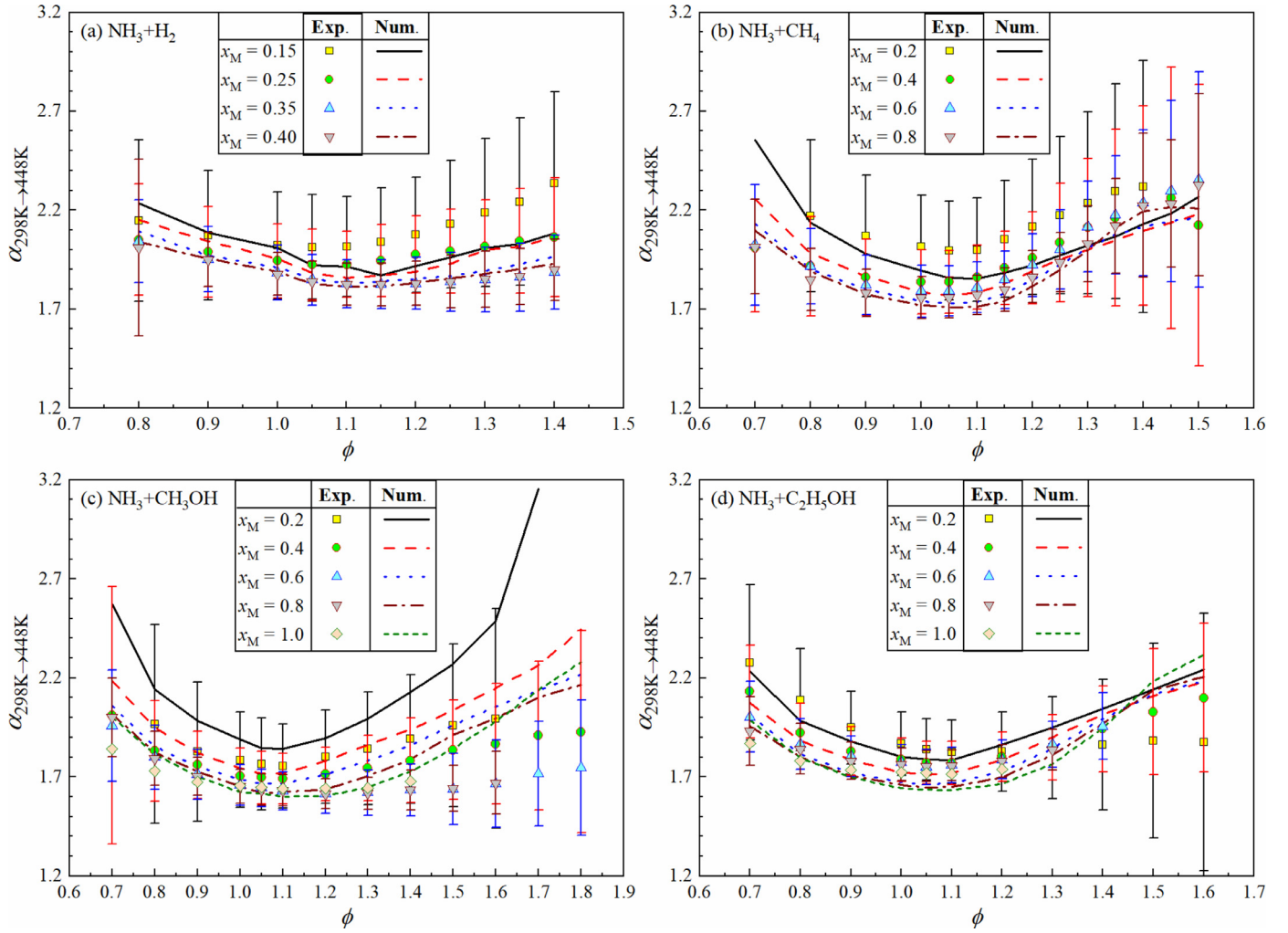


Fig. 6. Temperature coefficient of (a) NH_3+H_2 +air, (b) NH_3+CH_4 +air, (c) $\text{NH}_3+\text{CH}_3\text{OH}$ +air, and (d) $\text{NH}_3+\text{C}_2\text{H}_5\text{OH}$ +air flames, derived from the S_L at 298 K and 448 K, 1 atm.

difference in the experimentally and numerically derived α values points out underlying inconsistencies of the experimental data. Figure 6 shows the temperature coefficient α derived from the experimental and numerical data in Fig. 2 (298 K) and Fig. 5 (448 K) through Eq. (4)

$$\alpha_{T_u \rightarrow T_u^0} = \frac{\ln(S_L/S_L^0)}{\ln(T_u/T_u^0)} \quad (4)$$

Where the S_L and S_L^0 are the laminar burning velocities at T_u (448 K) and T_u^0 (298 K), respectively. The uncertainty of the experimentally derived α is evaluated from the uncertainties of the laminar burning velocities, as shown in Eq. (5) [3,54,55].

$$\Delta\alpha_{T_u \rightarrow T_u^0} = \frac{1}{\ln(T_u/T_u^0)} \left(\frac{\Delta S_L}{S_L} + \frac{\Delta S_L^0}{S_L^0} \right) \quad (5)$$

From Fig. 6, most of the experimental α values overlap with the simulation results within the uncertainty range, verifying good data consistency in the present experiment. For the NH_3+H_2 and lean to moderately rich $\text{NH}_3+\text{CH}_4/\text{CH}_3\text{OH}/\text{C}_2\text{H}_5\text{OH}$ mixtures, minimum α is always found at around stoichiometry, and the α coefficient decreases with increasing mixing ratio x_M . However, at the very rich side of the $\text{NH}_3+\text{CH}_4/\text{CH}_3\text{OH}/\text{C}_2\text{H}_5\text{OH}$ flames, the α values show non-monotonic dependence on x_M , and the shape of α vs. ϕ in some datasets are similar to that observed in the over-rich C1-C3 flames [55]. Since the over-rich flame structure has been

found in both hydrocarbon [55] and NH_3 [20] fueled flames, it is not surprising if it can also be found in their mixtures. The over-rich flame structure in [20,55] is characterized by, e.g., superadiabatic flame temperature (SAFT, the maximum flame temperature exceeding the adiabatic one) and changes in the overall activation energy E_a in adiabatic flames. Therefore, analyses were carried out based on these characteristics using adiabatic CEU-NH3 simulation results of the present fuel mixtures.

Figure 7 shows the overall activation energy E_a of $\text{NH}_3+\text{H}_2/\text{CH}_4/\text{CH}_3\text{OH}/\text{C}_2\text{H}_5\text{OH}$ +air the flames, as functions of equivalence ratio ϕ and mixing ratio x_M , where symbols denote the occurrence of SAFT in the corresponding conditions. The E_a data were derived from adiabatic flame simulation results using the CEU-NH3 mechanism through Eq. (6), based on the fact that the global flame structure doesn't change much in the limited unburnt temperature range investigated [54].

$$\alpha_{T_u^0 \rightarrow T_u} = \frac{E_a}{2R} \cdot X + x \quad (6)$$

In Eq. (6), X is related to the adiabatic flame temperature and x is a synthetic parameter related to reaction order. From Fig. 7, the E_a values tend to increase with larger x_M at lean to moderately rich sides of each mixture, but this is not always true due to the data scatterings. On the richer side, the E_a values have more irregular tendencies, while the equivalence ratio range of SAFT narrows at larger x_M , which could be anticipated by the performance in the NH_3 +air and M+air flames. The NH_3 +air flame has

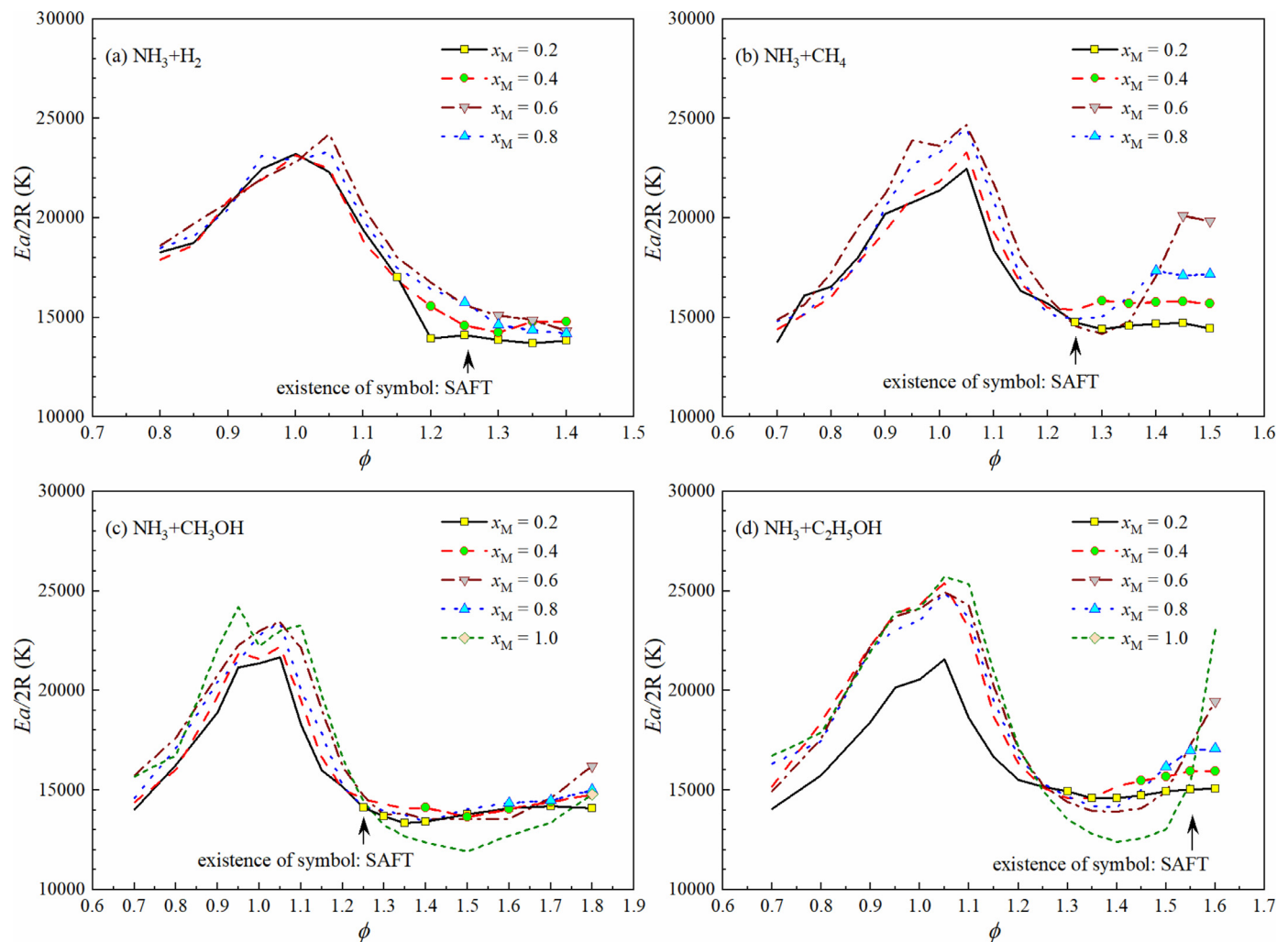


Fig. 7. Overall activation energy of (a) NH_3+H_2 +air, (b) NH_3+CH_4 +air, (c) $\text{NH}_3+\text{CH}_3\text{OH}$ +air, and (d) $\text{NH}_3+\text{C}_2\text{H}_5\text{OH}$ +air flames, derived from adiabatic simulation results.

a starting point of SAFT at $\phi = 1.1 - 1.15$, where its E_a has local minimum [20]; while the CH_4 +air flame has a starting point of SAFT at $\phi = 1.4$, where its E_a reaches local maximum [55]. Therefore, the starting point of SAFT in NH_3+CH_4 +air flames should be at somewhere between with the E_a being influenced both by the maximum and minimum points. Similar situations are there in the $\text{NH}_3+\text{CH}_3\text{OH}/\text{C}_2\text{H}_5\text{OH}$ flames. For the NH_3+H_2 flames, since H_2 +air flame has no over-rich flame structure [55], the adding of H_2 just dilutes the NH_3 fraction, postponing the occurrence of SAFT to the richer side of flames. The above discussion is also a supplementary proof that the interactions between NH_3 and M are not important, since otherwise there could be sudden changes of E_a and SAFT beginning points when the x_M changes.

5. Conclusion

In the present work, the laminar burning velocities of $\text{NH}_3+\text{H}_2/\text{CH}_4/\text{CH}_3\text{OH}/\text{C}_2\text{H}_5\text{OH} + \text{air}$ flames were measured using the heat flux method at 1 atm and 298 K to 448 K unburnt temperatures. Analyses based on the temperature coefficient α proves the consistency of the present experimental data, whereas analyses based on the normalized burning velocities ξ and energy fraction E_M at stoichiometry reveal that the NH_3 mixtures blended with additive M have the regulation $\xi = E_M^2$, no matter the M refers to H_2 , CH_4 , CH_3OH , or $\text{C}_2\text{H}_5\text{OH}$. Based on the above findings, it's

concluded that the blending of oxygenated alcohols into the NH_3 flame has a similar effect to the simple hydrocarbon fuels H_2 and CH_4 on the laminar burning velocity.

A new kinetic mechanism, named CEU-NH3, was developed and its simulation results show satisfactory agreement with the present experimental data and the literature data of ignition delay times and species profiles. This mechanism contains 91 species and 444 reactions, where 6 species and the corresponding 43 reactions could be deleted if the N_2O or NO_2 is not the main component in reactants. The present work validates this mechanism among the NH_3 , H_2 , CH_4 , CH_3OH , $\text{C}_2\text{H}_5\text{OH}$, and their blending mixtures, however, it has the potential to work on more complex systems with higher-order hydrocarbon fuels since species up to C4 have been inherited from the San Diego mechanism. By numerical analyses using the CEU-NH3 and the detailed Konnov mechanisms, the interactions between C- and N- containing species are found to be insignificant for the laminar burning velocities of the various NH_3 blending mixtures; while this is not true for the predictions of NH_3+CH_4 ignition delay times, where two C-N interactions reactions have been found with large A-factor sensitivities. This finding points out the possibility to further reduce the size of the CEU-NH3 mechanism, which could be helpful for the design and optimization of real combustors. All experimental data with uncertainties and the CEU-NH3 mechanism can be found in the Supplementary material.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the National Natural Science Foundation of China (51876192, 51621005) and the Fundamental Research Funds for the Central Universities (2020FZZX003-01-01).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2021.02.038](https://doi.org/10.1016/j.combustflame.2021.02.038).

References

- [1] E.C. Okafor, K.D.K.A. Somarathne, A. Hayakawa, T. Kudo, O. Kurata, N. Iki, H. Kobayashi, Towards the development of an efficient low-NOx ammonia combustor for a micro gas turbine, *Proc. Combust. Inst.* 37 (2019) 4597–4606.
- [2] O. Kurata, N. Iki, T. Inoue, T. Matsunuma, T. Tsujimura, H. Furutani, M. Kawano, K. Arai, E.C. Okafor, A. Hayakawa, H. Kobayashi, Development of a wide range-operable, rich-lean low-NOx combustor for NH₃ fuel gas-turbine power generation, *Proc. Combust. Inst.* 37 (2019) 4587–4595.
- [3] X. Han, Z. Wang, M. Costa, Z. Sun, Y. He, K. Cen, Experimental and kinetic modeling study of laminar burning velocities of NH₃/air, NH₃/H₂/air, NH₃/CO/air and NH₃/CH₄/air premixed flames, *Combust. Flame* 206 (2019) 214–226.
- [4] X. Han, Z. Wang, Y. He, Y. Zhu, K. Cen, Experimental and kinetic modeling study of laminar burning velocities of NH₃/syngas/air premixed flames, *Combust. Flame* 213 (2020) 1–13.
- [5] C.W. Gross, S.-C. Kong, Performance characteristics of a compression-ignition engine using direct-injection ammonia–DME mixtures, *Fuel* 103 (2013) 1069–1079.
- [6] A.J. Reiter, S.-C. Kong, Combustion and emissions characteristics of compression-ignition engine using dual ammonia–diesel fuel, *Fuel* 90 (2011) 87–97.
- [7] E.C. Okafor, Y. Naito, S. Colson, A. Ichikawa, T. Kudo, A. Hayakawa, H. Kobayashi, Experimental and numerical study of the laminar burning velocity of CH₄–NH₃–air premixed flames, *Combust. Flame* 187 (2018) 185–198.
- [8] P. Dirrenberger, P.-A. Glaude, R. Bounaceur, H. Le Gall, A.P. Da Cruz, A. Konnov, F.J.F. Battin-Leclerc, Laminar burning velocity of gasolines with addition of ethanol, *Fuel* 115 (2014) 162–169.
- [9] L. Sileghem, V. Alekseev, J. Vancouillie, E. Nilsson, S. Verhelst, A.J.F. Konnov, Laminar burning velocities of primary reference fuels and simple alcohols, 115 (2014) 32–40.
- [10] O. Mathieu, E.L. Petersen, Experimental and modeling study on the high-temperature oxidation of Ammonia and related NOx chemistry, *Combust. Flame* 162 (2015) 554–570.
- [11] Y. Song, H. Hashemi, J.M. Christensen, C. Zou, P. Marshall, P. Glarborg, Ammonia oxidation at high pressure and intermediate temperatures, *Fuel* 181 (2016) 358–365.
- [12] H. Nakamura, S. Hasegawa, T. Tezuka, Kinetic modeling of ammonia/air weak flames in a micro flow reactor with a controlled temperature profile, *Combust. Flame* 185 (2017) 16–27.
- [13] J. Otomo, M. Koshi, T. Mitsumori, H. Iwasaki, K. Yamada, Chemical kinetic modeling of ammonia oxidation with improved reaction mechanism for ammonia/air and ammonia/hydrogen/air combustion, *Int. J. Hydrogen Energ.* 43 (2018) 3004–3014.
- [14] Z. Tian, Y. Li, L. Zhang, P. Glarborg, F. Qi, An experimental and kinetic modeling study of premixed NH₃/CH₄/O₂/Ar flames at low pressure, *Combust. Flame* 156 (2009) 1413–1426.
- [15] K.P. Shrestha, L. Seidel, T. Zeuch, F. Mauss, Detailed kinetic mechanism for the oxidation of ammonia including the formation and reduction of nitrogen oxides, *Energy Fuels* 32 (2018) 10202–10217.
- [16] U. Mechanism, Chemical-kinetic mechanisms for combustion applications, Mechanical and Aerospace Engineering (Combustion Research), University of California at San Diego, <http://web.eng.ucsd.edu/mae/groups/combustion/mechanism.html>, (2014).
- [17] H. Xiao, S. Lai, A. Valera-Medina, J. Li, J. Liu, H. Fu, Experimental and modeling study on ignition delay of ammonia/methane fuels, 44 (2020) 6939–6949.
- [18] V.A. Alekseev, A.A. Konnov, Data consistency of the burning velocity measurements using the heat flux method: hydrogen flames, *Combust. Flame* 194 (2018) 28–36.
- [19] A.A. Konnov, A. Mohammad, V.R. Kishore, N.I. Kim, C. Prathap, S. Kumar, A comprehensive review of measurements and data analysis of laminar burning velocities for various fuel+ air mixtures, *Prog. Energy Combust. Sci.* 68 (2018) 197–267.
- [20] X. Han, Z. Wang, Y. He, Y. Liu, Y. Zhu, A.A. Konnov, The temperature dependence of the laminar burning velocity and superadiabatic flame temperature phenomenon for NH₃/air flames, *Combust. Flame* 217 (2020) 314–320.
- [21] X. Han, Z. Wang, Y. He, S. Wang, Y. Liu, A.A. Konnov, Temperature dependence of the laminar burning velocity for n-heptane and iso-octane/air flames, *Fuel* 276 (2020) 118007.
- [22] Z. Wang, X. Han, Y. He, S. Wang, R. Ji, Y. Zhu, K. Cen, Investigation of flame and burner plate interaction during the heat flux method used for laminar burning velocity measurement, *Fuel* 266 (2020) 117051.
- [23] V.A. Alekseev, J.D. Naucier, M. Christensen, E.J.K. Nilsson, E.N. Volkov, L.P.H. de Goey, A.A. Konnov, Experimental uncertainties of the heat flux method for measuring burning velocities, *Combust. Sci. Technol.* 188 (2016) 853–894.
- [24] K. Bosschaart, L. De Goey, The laminar burning velocity of flames propagating in mixtures of hydrocarbons and air measured with the heat flux method, *Combust. Flame* 136 (2004) 261–269.
- [25] K.P. Shrestha, C. Lhuillier, A.A. Barbosa, P. Brequigny, F. Contino, C. Mounaïm-Rousselle, L. Seidel, F. Mauss, An experimental and modeling study of ammonia with enriched oxygen content and ammonia/hydrogen laminar flame speed at elevated pressure and temperature, *Proceedings of the Combustion Institute* (2020), doi:[10.1016/j.proci.2020.06.197](https://doi.org/10.1016/j.proci.2020.06.197).
- [26] P. Glarborg, J.A. Miller, B. Ruscic, S.J. Klippenstein, Modeling nitrogen chemistry in combustion, *Prog. Energy Combust. Sci.* 67 (2018) 31–68.
- [27] Y. Song, L. Marrodán, N. Vin, O. Herbinet, E. Assaf, C. Fittschen, A. Stagni, T. Faravelli, M.U. Alzueta, F. Battin-Leclerc, The sensitizing effects of NO₂ and NO on methane low temperature oxidation in a jet stirred reactor, *Proceedings of the Combustion Institute*, 37 (2019), pp. 667–675.
- [28] X. Han, M. Lubrano Lavadera, C. Brackmann, Z. Wang, Y. He, A.A. Konnov, Experimental and kinetic modeling study of NO formation in premixed CH₄+O₂+N₂ flames, *Combust. Flame* 223 (2021) 349–360.
- [29] H. Nakamura, M. Shindo, Effects of radiation heat loss on laminar premixed ammonia/air flames, *Proc. Combust. Inst.* 37 (2019) 1741–1748.
- [30] T. Varga, C. Olm, T. Nagy, I.G. Zsély, É. Valkó, R. Pálvölgyi, H.J. Curran, T. Turányi, Development of a joint hydrogen and syngas combustion mechanism based on an optimization approach, *Int. J. Chem. Kinetics* 48 (2016) 407–422.
- [31] J.C. Prince, C. Treviño, F.A. Williams, A reduced reaction mechanism for the combustion of n-butane, *Combust. Flame* 175 (2017) 27–33.
- [32] C. Brackmann, T. Methling, M. Lubrano Lavadera, G. Capriolo, A.A. Konnov, Experimental and modeling study of nitric oxide formation in premixed methanol + air flames, *Combust. Flame* 213 (2020) 322–330.
- [33] K. Devriendt, M. Vanpoppel, W. Boullart, J. Peeters, Kinetic investigation of the CH₂ ((X) over-Tilde (3) B (1)) + h- CH (X (2) Pi) + h-2 reaction in the temperature-range 400K-less-than-T-less-than-1000K, *J. Phys. Chem.* 99 (1995) 16953–16959.
- [34] C. Mehlmann, M.J. Frost, D.E. Heard, B.J. Orr, P. Nelson, *Faraday Transactions*, Rate constants for removal of CH (D)(v= 0 and 1) by collisions with N₂, CO, O₂, NO and NO₂ at 298K and with CO₂ at 296< T/K<=, 873, 92 (1996) 2335–2341.
- [35] D. Baulch, C.T. Bowman, C. Cobos, R. Cox, T. Just, J. Kerr, M. Pilling, D. Stocker, J. Troe, W. Tsang, Evaluated kinetic data for combustion modeling: supplement II, *J. Phys. Chem. Ref. Data* 34 (2005) 757–1397.
- [36] M. Röhrig, E.L. Petersen, D.F. Davidson, R.K. Hanson, C.T. Bowman, Measurement of the rate coefficient of the reaction CH+O₂ → products in the temperature range 2200 to 2600K, *Int. J. Chem. Kinet.* 29 (1997) 781–789.
- [37] G. Smith, Y. Tao, H. Wang, Foundational fuel chemistry model version 1.0 (FFCM-1) (2020).
- [38] A.W. Jasper, S.J. Klippenstein, L.B. Harding, B. Ruscic, Kinetics of the reaction of methyl radical with hydroxyl radical and methanol decomposition, *J. Phys. Chem. A* 111 (2007) 3932–3950.
- [39] E.E. Dames, D.M. Golden, Master equation modeling of the unimolecular decompositions of hydroxymethyl (CH₂OH) and methoxy (CH₃O) radicals to formaldehyde (CH₂O) + H, *J. Phys. Chem. A* 117 (2013) 7686–7696.
- [40] E.E. Dames, Master equation modeling of the unimolecular decompositions of α-hydroxyethyl (CH₃CHOH) and ethoxy (CH₃CH₂O) radicals, *Int. J. Chem. Kinet.* 46 (2014) 176–188.
- [41] V.A. Alekseev, S.S. Matveev, I.V. Chechet, S.G. Matveev, A.A. Konnov, Laminar burning velocities of methylcyclohexane + air flames at room and elevated temperatures: a comparative study, *Combust. Flame* 196 (2018) 99–107.
- [42] A. Mohammad, K.A. Juhany, Laminar burning velocity and flame structure of DME/methane + air mixtures at elevated temperatures, *Fuel* 245 (2019) 105–114.
- [43] Y. Wu, V. Modica, B. Rossow, F. Grisch, Effects of pressure and preheating temperature on the laminar flame speed of methane/air and acetone/air mixtures, *Fuel* 185 (2016) 577–588.
- [44] E. Hu, X. Li, X. Meng, Y. Chen, Y. Cheng, Y. Xie, Z. Huang, Laminar flame speeds and ignition delay times of methane–air mixtures at elevated temperatures and pressures, *Fuel* 158 (2015) 1–10.
- [45] L. Sileghem, V.A. Alekseev, J. Vancouillie, E.J.K. Nilsson, S. Verhelst, A.A. Konnov, Laminar burning velocities of primary reference fuels and simple alcohols, *Fuel* 115 (2014) 32–40.
- [46] J. Vancouillie, M. Christensen, E.J.K. Nilsson, S. Verhelst, A.A. Konnov, Temperature dependence of the laminar burning velocity of methanol flames, *Energy Fuels* 26 (2012) 1557–1564.
- [47] A. Katoch, M. Asad, S. Minaev, S. Kumar, Measurement of laminar burning velocities of methanol–air mixtures at elevated temperatures, *Fuel* 182 (2016) 57–63.

- [48] F. Rau, S. Hartl, S. Voss, M. Still, C. Hasse, D. Trimis, Laminar burning velocity measurements using the Heat Flux method and numerical predictions of iso-octane/ethanol blends for different preheat temperatures, *Fuel* 140 (2015) 10–16.
- [49] A. Katoch, A. Millán-Merino, S. Kumar, Measurement of laminar burning velocity of ethanol-air mixtures at elevated temperatures, *Fuel* 231 (2018) 37–44.
- [50] H. El Merhubi, A. Kéromnès, G. Catalano, B. Lefort, L.L. Moyne, A high pressure experimental and numerical study of methane ignition, *Fuel* 177 (2016) 164–172.
- [51] K.E. Noorani, B. Akih-Kumgeh, J.M. Bergthorson, Comparative high temperature shock tube ignition of C1–C4 primary alcohols, *Energy Fuels* 24 (2010) 5834–5843.
- [52] G.L. Dugger, D.D. Graab, Flame velocities of hydrocarbon-oxygen-nitrogen mixtures, *Symp. (Int.) Combust.* 4 (1953) 302–310.
- [53] G.L. Dugger, D.M. Simon, Prediction of flame velocities of hydrocarbon flames, *Symp. (Int.) Combust.* 4 (1953) 336–345.
- [54] X. Han, Z. Wang, S. Wang, R. Whiddon, Y. He, Y. Lv, A.A. Konnov, Parametrization of the temperature dependence of laminar burning velocity for methane and ethane flames, *Fuel* 239 (2019) 1028–1037.
- [55] X. Han, Z. Wang, Y. He, S. Wang, Y. Zhu, A.A. Konnov, Over-rich combustion of CH₄, C₂H₆, and C₃H₈ +air premixed flames investigated by the heat flux method and kinetic modeling, *Combust. Flame* 210 (2019) 339–349.